

Learning Activity Sheet for General Mathematics

Quarter 10

Unit

11

Learning Activity Sheet for Mathematics Grade 11
Quarter 4: Unit 11

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LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	4
Lesson No.:	11.1	Date:	
Lesson Title/ Topic:	Null and Alternative Hypothesis		
Name:		Grade & Section:	

I. Activity No. 1	Practicing Learned Skills
II. Objectives:	At the end of the activity, you will be able apply your understanding on null and alternative hypothesis.
III. Materials Needed:	pen and paper
IV. Instructions:	Do as indicated. Part 1: Let's Try Together. 1. Read each scenario carefully. 2. Identify the Null Hypothesis (H_0) – statement of no effect/difference. 3. Identify the Alternative Hypothesis (H_1) – statement that there is an effect/difference. Scenarios: 1. A company claims its new fertilizer increases corn yield. 2. A teacher says her new math technique improves problem-solving skills. 3. A hospital tests whether a new patient scheduling system reduces wait times. 4. A gym claims its new workout program improves stamina in 4 weeks. Part 2: State Your Hypotheses Instructions: Write H_0 and H_1 for each scenario. 1. A doctor says a new drug lowers blood pressure better than the current one. 2. A coffee shop claims its coffee keeps people awake longer. 3. A company introduces a battery it claims lasts 20% longer. 4. A school believes athletes have higher GPAs than non-athletes. 5. A clothing brand claims its new fabric is more durable. 6. A university claims its online learning platform improves learner retention. 7. A tech company tests whether its new app improves user satisfaction. 8. A city claims that installing more streetlights reduces crime rates. 9. A nutritionist says eating almonds daily improves concentration. 10. A bookstore claims its loyalty program increases monthly sales.
V. Synthesis:	Answer the following questions: 1. How do we know if a statement should be H_0 or H_1? 2. What happens if H_0 is rejected but was actually true?

I. Activity No. 2	Making Real-World Connections
II. Objectives:	At the end of the activity, you will be able apply your understanding on null and alternative hypothesis.
III. Materials Needed:	pen and paper
IV. Instructions:	Perform the indicated tasks. Option 1: School-Based Research Idea Instructions: 1. Think of a School-Based Issue • Choose a topic related to school life, learning, or learner behavior. • The question should be measurable and realistic. 2. Write Your Research Question

- Phrase it as a clear, testable question.
- Focus on how one factor may affect another.

3. Write the Null Hypothesis (H_0)

- State that the factor you are testing has no effect on the outcome.
- Use neutral wording.

4. Write the Alternative Hypothesis (H_1)

- State that the factor does affect, improve, or reduce the outcome.
- This should align with your research question.

5. Keep Statements Clear and Simple

- Hypotheses must be written as statements, not questions.
- Do not include explanations or results.

Task:

Learners propose a researchable school problem and state H_0 and H_1 .

Example: “Does using background music while studying affect test scores?”

H_0 : Background music has no effect on test scores.

H_1 : Background music improves test scores.

Option 2: Community Application

Instructions:

1. Form a Group

- Work with your assigned group members.

2. Identify a Community Issue

- Choose a problem related to:
 - Health
 - Environment
 - Education
 - Business
 - Public safety
- The issue should be local, realistic, and measurable.

3. Create a Research Question

- Write a question that investigates whether an action or program affects a community outcome.
- The question should be testable using data.

4. Write the Null Hypothesis (H_0)

- State that the action or program has no effect on the issue.
- Use clear and neutral wording.

5. Write the Alternative Hypothesis (H_1)

- State that the action or program does affect or improve the issue.
- This should directly match your research question.

6. Prepare to Share

- Be ready to explain your question and hypotheses to the class.

Task: Groups identify a community issue (e.g., health, environment, business) and create hypotheses.

Example: “Does recycling awareness reduce waste in the barangay?”

H_0 : Recycling awareness has no effect on waste reduction.

H_1 : Recycling awareness reduces waste.

Option 3: Mini-Case Studies

Instructions:

1. Read the Mini-Case Carefully

- Identify what is being introduced, tested, or compared.

	- Determine the outcome being measured. - Identify the Variables - What is the factor or treatment? - What is the result or outcome? - Write the Null Hypothesis (H_0) - State that the factor being tested has no effect on the outcome. - Use clear, neutral wording. - Write the Alternative Hypothesis (H_1) - State that the factor does affect or improve the outcome. - Match the claim implied in the case. - Write Hypotheses as Statements - Do not use questions. - Keep answers short and direct. Task: Teacher provides short case studies → learners identify hypotheses. Example: A farmer applies organic fertilizer on half of his rice fields to test yield difference. H_0: Organic fertilizer has no effect on rice yield. H_1: Organic fertilizer increases rice yield.
V. Synthesis:	Answer the following questions: - How are the null and alternative hypotheses similar or different when you apply them to a school-based issue versus a community problem? Give an example for each. - Choose either a school-based or community issue and outline a mini-case study that tests your research question. Clearly state the variables, H_0, and H_1, and explain why your hypotheses make sense. - Look at your hypotheses from any activity. How could confirming or rejecting H_0 affect decision-making in school or in the community? Provide a specific example. - Select one factor (e.g., study habits, recycling programs, fertilizer use) and create three different hypotheses—one for a school-based study, one for a community application, and one for a mini-case study. Explain the differences in wording and context. - Based on your experience with all three activities, how would you revise a poorly written hypothesis to make it testable and meaningful in a real-world scenario? Provide before-and-after examples.

I. Activity No. 3	Assessing Learning Outcomes
II. Objectives:	At the end of the activity, you will be able apply your understanding on null and alternative hypothesis.
III. Materials Needed:	pen and paper
IV. Instructions:	Do as indicated. A. Choose the Correct Answer - The null hypothesis (H_0) usually states: - There is an effect or difference - There is no effect or difference - Which of the following is an example of an alternative hypothesis (H_1)? “The new drug has no effect on headaches.” “The new drug reduces headaches better than the old one.” - Which statement best describes the role of H_1? - It represents what researchers want to disprove - It represents the claim or effect being tested - In hypothesis testing, researchers usually try to: - Prove H_0 is true - Find evidence against H_0 - Which pair of hypotheses is correctly stated?

	a. a) H_0: There is a difference; H_1: There is no difference b. b) H_0: There is no difference; H_1: There is a difference B. Identify Whether the Statement is H_0 or H_1 1. "The new teaching method has no effect on learner performance." 2. "Athletes have higher GPAs than non-athletes." 3. "There is no significant difference in test scores between online and face-to-face classes." 4. "Learners who sleep at least 8 hours perform better on exams." 5. "The new phone battery does not last longer than the previous model." C. Constructing Hypotheses 1. Create one research question of your own related to school, health, or daily life. 2. Write the corresponding: - o H_0 (null hypothesis) - o H_1 (alternative hypothesis)
V. Synthesis:	Answer in 2–3 sentences each. 1. Many learners think H_0 is always false. Why is this incorrect, and what is the real purpose of H_0? 2. What difficulties did you encounter when deciding whether a statement belongs to H_0 or H_1? 3. Why is it important to write hypotheses carefully before conducting any statistical test?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	4
Lesson No.:	11.2	Date:	
Lesson Title/ Topic:	Type I and Type II Error		
Name:		Grade & Section:	

I. Activity No. 1	Practicing Learned Skills
II. Objectives:	At the end of the activity, you will able to apply your understanding of Type I and Type II errors and the level of significance (α).
III. Materials Needed:	pen and paper
IV. Instructions:	Do as indicated. Part 1: Error Analysis Quiz Directions: Read each question carefully and choose the correct answer. For each, explain why your choice is correct. - In a clinical trial, the researchers reject the null hypothesis when the new drug actually has no effect. This is an example of: - Type I Error - Type II Error - Correct Decision - A smoke detector fails to sound during a fire. What type of error is this? - Type I Error - Type II Error - Correct Decision - A test with $\alpha = 0.05$ means: - There is a 5% chance of making a Type II error - There is a 5% chance of making a Type I error - There is a 95% chance of a Type I error - What happens if the significance level α is set very low (e.g., 0.01)? - Increased chance of Type I error - Increased chance of Type II error - No change in errors - When the consequences of a false positive are serious, you should: - Increase α to 0.10 - Decrease α to 0.01 - Ignore α Part 2: Scenario-Based Decision Making Directions: For each scenario below: - Identify the null and alternative hypotheses. - Determine which type of error is more serious. - Recommend an appropriate α level and explain your choice. Scenarios: - Testing if a new airplane model is safer than the current model. - Screening job applicants for criminal records. - Detecting a disease outbreak in a community. Part 3: Group Debate Directions:

	Divide the class into two groups. One group argues that Type I errors are more serious, the other that Type II errors are more serious. Each group must: • Provide real-world examples. • Explain consequences. • Discuss how significance level affects their position. After the debate, conduct a reflection discussing that seriousness depends on context.
V. Synthesis:	Answer the following questions: 1. In which situations is a Type I error more serious? Give real-life examples. 2. In which situations is a Type II error more serious? Give real-life examples. 3. How does the level of significance (α) influence the balance between Type I and Type II errors? 4. After completing these activities, how confident are you in identifying errors in a hypothesis test? 5. How can understanding these errors help you make better decisions in daily life or in scientific research?

I. Activity No. 2	Making Real-World Connections														
II. Objectives:	At the end of the activity, you will able to apply your understanding of Type I and Type II errors and the level of significance (α).														
III. Materials Needed:	pen and paper														
IV. Instructions:	Follow the steps. Step 1: Understand Your Variable 1. Review your assigned variable from the performance task (e.g., hours of study, sleep hours, practice tests). 2. Think about how this variable might affect academic performance. Step 2: Formulate Hypotheses 1. Write a null hypothesis (H_0): Assume the variable has no effect on academic performance. 2. Write an alternative hypothesis (H_1): Assume the variable does have an effect. Step 3: Identify Possible Errors 1. Describe a Type I error for your variable: What would happen if you wrongly conclude there is an effect when there isn't? 2. Describe a Type II error for your variable: What would happen if you fail to detect an effect when there actually is one? Step 4: Decide on Level of Significance (α) 1. Consider the consequences of each error. 2. Choose an α level (0.01, 0.05, or 0.10) that balances caution and risk. 3. Justify your choice in writing. Step 5: Organize Your Ideas 1. Create a simple table to summarize your variable, hypotheses, errors, and α choice. Example: <table><tr><th>Variable</th><th>H_0</th><th>H_1</th><th>Type I Error</th><th>Type II Error</th><th>α</th><th>Reason</th></tr><tr><td>Study Hours</td><td>No effect on grades</td><td>Has effect</td><td>Conclude effect when none exists</td><td>Fail to detect effect</td><td>0.05</td><td>Moderate risk; false positives may mislead strategies</td></tr></table>	Variable	H_0	H_1	Type I Error	Type II Error	α	Reason	Study Hours	No effect on grades	Has effect	Conclude effect when none exists	Fail to detect effect	0.05	Moderate risk; false positives may mislead strategies
Variable	H_0	H_1	Type I Error	Type II Error	α	Reason									
Study Hours	No effect on grades	Has effect	Conclude effect when none exists	Fail to detect effect	0.05	Moderate risk; false positives may mislead strategies									
V. Synthesis:	Answer the following questions: 1. Which error is more serious for your variable? 2. How does your α choice influence the risk of making errors?														

I. Activity No. 3	Assessing Learning Outcomes
II. Objectives:	At the end of the activity, you will be able to apply your understanding of Type I and Type II errors and the level of significance (α).
III. Materials Needed:	pen and paper

IV. Instructions:	Perform the indicated tasks. Part 1. Multiple-Choice Directions: Choose the correct answer. 1. A significance level (α) of 0.01 means: a. There is a 1% chance of making a Type II error b. There is a 1% chance of making a Type I error c. There is a 99% chance of making a Type I error 2. Rejecting H_0 when it is actually true is called: a. Type I Error b. Type II Error c. Correct Decision 3. Failing to reject H_0 when it is actually false is called: a. Type I Error b. Type II Error c. Correct Decision 4. Which scenario requires a very low α level? a. Choosing a flavor for ice cream b. Approving a new life-saving drug c. Choosing a class activity 5. Increasing α from 0.01 to 0.10 generally: a. Increases Type I error and decreases Type II error b. Decreases both Type I and Type II errors c. Increases Type II error and decreases Type I error Part 2. Scenario-Based Short Answer Directions: For each scenario: 1. Identify H_0 and H_1 2. State which error (Type I or II) is more serious 3. Suggest an appropriate α level and justify Scenarios: 1. A smoke detector in a school detects fire. 2. A company tests a new software for security vulnerabilities. 3. A hospital tests a new vaccine for safety. 4. Airport security screens luggage for explosives. 5. A plagiarism detection tool checks learner essays. Part 3. Matching Activity Directions: Match the scenarios to the correct error type. Scenarios: A. A pregnancy test says pregnant, but the woman is not. B. A fire alarm does not go off during a real fire. C. A COVID-19 test correctly identifies infection. D. Security alerts police with no intruder. E. A learner cheats but the teacher does not notice. Error Types: 1. Type I Error 2. Type II Error 3. Correct Decision
V. Synthesis:	Answer the following questions: 1. How does choosing a higher or lower α affect the likelihood of Type I and Type II errors? 2. Can you think of a real-life situation where a Type I error would be more serious than a Type II error, and vice versa? 3. Based on these activities, what general rules can you draw about deciding on an appropriate significance level in different contexts.

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	4
Lesson No.:	11.3	Date:	
Lesson Title/ Topic:	Test of Mean for One Population		
Name:		Grade & Section:	

I. Activity No. 1	Practicing Learned Skills
II. Objectives:	At the end of the activity, students will be able to apply appropriate hypothesis testing procedures (z-test or t-test) to evaluate claims about population means, interpret statistical results, and draw conclusions using real-life scenarios.
III. Materials Needed:	pen and paper
IV. Instructions:	Read and solve each problem. 1. To promote digital wellness, the school recommends that learners limit their recreational screen time to no more than 2 hours per day. Mr. Malinaw, a learner researcher, surveys 100 learners and finds that the average screen time is 2.3 hours, with a sample standard deviation of 0.8 hours. Using a significance level of 0.05, can we conclude that learners are spending more time than recommended? 2. A school wellness committee of Maalaga National High School recommends that learners get at least 7 hours of sleep per night, even during exam week. A learner- researcher surveys 15 classmates and finds that the average sleep duration is 6.4 hours, with a sample standard deviation of 0.8 hours. Using a significance level of 0.05, can we conclude that learners are sleeping less than the recommended amount? 3. The Mabusog-lusog National High School cafeteria claims that their budget meal contains an average of 500 calories. A nutritionist believes the meals contain more than this amount. From a previous nationwide study, the population standard deviation is known to be 25 calories. The nutritionist analyzes a sample of 100 meals and finds an average of 506 calories. Using a 0.05 significance level, is there enough evidence to support the claim of the nutritionist? 4. Sama-sama National High School promotes shared meals at home, recommending 5 meals per week with parents. Suppose Ms. Dimagibagiba, a learner researcher, surveys 8 learners and finds that the average is 6.1 shared meals, with a sample standard deviation of 1.0 meal. Using a significance level of 0.05, can we conclude that learners are sharing more meals than recommended?
V. Synthesis:	Answer the following questions: - Examine all four problems and identify which use a z-test and which use a t-test. Explain why the choice of test differs across problems, considering factors such as sample size and whether the population standard deviation is known. - After performing the hypothesis tests, discuss how the conclusions for each scenario could impact decision-making in school or community settings. For example, how could findings about screen time, sleep, calories, or shared meals inform policies, programs, or recommendations? - Imagine you want to investigate another health or wellness behavior of students (e.g., daily water intake, study hours, or exercise frequency). Formulate a research question, state the null and alternative hypotheses, decide which statistical test you would use,

	and justify why it is appropriate based on the sample size and data available.
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I. Activity No. 2	Making Real-World Connections
II. Objectives:	At the end of the activity, students will be able to apply appropriate hypothesis testing procedures (z-test or t-test) to evaluate claims about population means, interpret statistical results, and draw conclusions using real-life scenarios.
III. Materials Needed:	pen and paper
IV. Instructions:	Read each problem and perform the indicated tasks. <i>Problem:</i> A consumer advocacy group in a certain province is concerned that residents may be paying more for electricity than the national average of ₱2,500 per month. To investigate, they randomly survey 50 households and record their monthly electricity bills. The population standard deviation is known to be ₱400. The sample yields an average monthly bill of ₱2,580. <i>Question:</i> At a 5% significance level, is there enough evidence to conclude that the average monthly electricity bill exceeds the national average? Gaming Time: Just Right or Too Much? <i>Problem:</i> Suppose Masunurin National High School conducts a study on learners' digital habits. Ma'am Maaro, school guidance counselor, is concerned that learners may be spending more time playing mobile games than the recommended maximum of 2 hours per day. She randomly selects a sample of 25 Grade 12 learners and records their average daily gaming time during a regular school week. The results show: <i>Sample mean: 2.4 hours</i> <i>Sample standard deviation: 0.6 hours</i> <i>Sample size: $n = 25$</i> <i>Question:</i> Assuming the population of learners has a normal distribution, test at the 5% significance level whether learners are spending more than 2 hours per day on mobile games.
V. Synthesis:	Answer the following questions: - Both problems involve testing whether a sample mean exceeds a known value, but the first problem uses a population standard deviation while the second uses a sample standard deviation. How does this difference affect the choice of statistical test (z-test vs. t-test)? Explain your reasoning with reference to both scenarios. - After performing the hypothesis tests for both the electricity bills and gaming time problems, discuss how rejecting or failing to reject the null hypothesis could influence decision-making for the province or the school. How might the results inform policy, recommendations, or interventions? - Imagine you want to investigate another behavior or expense (e.g., daily social media use or monthly water bills) using the same approach. Formulate a research question, define the null and alternative hypotheses, decide which statistical test you would use, and explain why it is appropriate.

I. Activity No. 3	Assessing Learning Outcomes
II. Objectives:	At the end of the activity, students will be able to apply appropriate hypothesis testing procedures (z-test or t-test) to evaluate claims about population means, interpret statistical results, and draw conclusions using real-life scenarios.

III. Materials Needed:	pen and paper
IV. Instructions:	Solve the following problems as required. Show a complete solution to get complete points. 1. A disaster response team claims that each relief pack weighs exactly 5 kilograms. A volunteer measures a random sample of 40 relief packs and finds an average weight of 5.1 kg. The population standard deviation is known to be 0.3 kg. At a 5% significance level, is there enough evidence to conclude that the average weight of the relief packs is different from the claimed 5 kg? 2. A teacher believes learners sleep less than 8 hours during exam week. A sample of 20 STEM learners shows an average of 7.3 hours, with a sample standard deviation of 0.7 hours. At a 5% significance level, is the average sleep time significantly less than 8 hours? 3. A government agency claims citizens complete a feedback form in 10 minutes or less. A sample of 100 responses shows an average of 10.4 minutes, with a known population standard deviation of 2.5 minutes. At a 5% significance level, is the average completion time greater than 10 minutes?
V. Synthesis:	Answer the following questions: 1. How can hypothesis testing help validate or challenge claims in each situation? 2. How does the choice between a z-test and a t-test affect the reliability of your conclusion in each scenario? 3. If we fail to reject the null hypothesis or find no significant difference, can we conclude that the original claim is always true? Explain your answer.

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	4
Lesson No.:	11.4	Date:	
Lesson Title/ Topic:	Test of Means for Two Populations (Paired and Unpaired)		
Name:		Grade & Section:	

I. Activity No. 1	Practicing Learned Skills																																											
II. Objectives:	At the end of the activity, students will be able to apply the appropriate hypothesis test for two independent sample means (z-test or t-test), determine whether a statistically significant difference exists between the groups, and draw real-world conclusions based on the findings.																																											
III. Materials Needed:	pen and paper																																											
IV. Instructions:	Read and solve each problem. 1. As part of a community-based outreach in Barangay Maliwanag, learners are tasked with evaluating battery performance for use in emergency kits. Two brands of battery, Meh Brand and Moh Brand, are being considered for distribution during typhoon season. To help the barangay council make an informed decision, learners test 10 batteries from each brand and record their lifespans (in hours): <table><tr><td>Meh Brand:</td><td>88</td><td>85</td><td>89</td><td>80</td><td>81</td><td>72</td><td>90</td><td>85</td><td>84</td><td>91</td></tr><tr><td>Moh Brand:</td><td>80</td><td>79</td><td>77</td><td>82</td><td>75</td><td>81</td><td>77</td><td>73</td><td>85</td><td>83</td></tr></table> Assuming that the conditions for inference are satisfied, which statistical test is most appropriate to determine whether Meh Brand batteries have a longer average lifespan than Moh Brand batteries? 2. At Maalalahanin National High School, ten senior high learners are quietly struggling. With deadlines piling up, expectations rising, and personal worries weighing heavily, they report high levels of academic stress. The school’s guidance office, led by Ma’am Mahinahon, launches a peer counseling circle, a safe space where learners can share, listen, and support one another. The program runs for two weeks. Each session begins with a short reflection, a prayer, and a prompt like “What’s one thing you’re carrying today?” Learners learn to name their emotions, validate each other’s experiences, and practice grounding techniques. To assess the impact, Ma’am Mahinahon asks each learner to rate their anxiety level on a scale of 1–10 before and after the program. The results are as follows: <table><tr><td>Learner</td><td>Before</td><td>After</td></tr><tr><td>1</td><td>8</td><td>6</td></tr><tr><td>2</td><td>7</td><td>5</td></tr><tr><td>3</td><td>9</td><td>6</td></tr><tr><td>4</td><td>6</td><td>4</td></tr><tr><td>5</td><td>8</td><td>5</td></tr><tr><td>6</td><td>7</td><td>6</td></tr></table>	Meh Brand:	88	85	89	80	81	72	90	85	84	91	Moh Brand:	80	79	77	82	75	81	77	73	85	83	Learner	Before	After	1	8	6	2	7	5	3	9	6	4	6	4	5	8	5	6	7	6
Meh Brand:	88	85	89	80	81	72	90	85	84	91																																		
Moh Brand:	80	79	77	82	75	81	77	73	85	83																																		
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	7	9	7											
	8	6	5											
	9	8	6											
	10	7	5											
	Is there statistical evidence that the peer counseling circle reduced learners' anxiety levels? Use a 5 level of significance. 3. Two barangays, Barangay Malusog and Barangay Masigla launch distinct school feeding programs to address learner nutrition and learning readiness. Both programs run for two months, offering daily meals designed to boost energy, focus, and overall health. Barangay Malusog partners with local farmers to serve fresh vegetable-based meals. Barangay Masigla collaborates with a nutrition NGO to provide fortified rice and protein-rich snacks. After the program, learners from each barangay take a standardized nutrition assessment scored out of 100. Health officers randomly sample 40 learners from each barangay and record the following: <table> <tr> <th>Barangay</th><th>Mean score</th><th>Population standard deviation</th><th>Sample size</th></tr> <tr> <td>Malusog</td><td>85</td><td>5</td><td>40</td></tr> <tr> <td>Masigla</td><td>82</td><td>6</td><td>40</td></tr> </table> Is there statistical evidence that Barangay Malusog's feeding program is more effective? Use a 5% level of significance, assuming known population standard deviations.			Barangay	Mean score	Population standard deviation	Sample size	Malusog	85	5	40	Masigla	82	6
Barangay	Mean score	Population standard deviation	Sample size											
Malusog	85	5	40											
Masigla	82	6	40											
V. Synthesis:	Answer the following questions: 1. What did you learn from your activity? 2. What made the problem challenging or meaningful? 3. How would you explain your solution to someone in a lower tier?													

I. Activity No. 2	Making Real-World Connections												
II. Objectives:	At the end of the activity, students will be able to apply the appropriate hypothesis test for two independent sample means (z-test or t-test), determine whether a statistically significant difference exists between the groups, and draw real-world conclusions based on the findings.												
III. Materials Needed:	pen and paper												
IV. Instructions:	Read and analyze each scenario and perform the indicated task. <i>Scenario:</i> At Maaral National High School, the senior high department is evaluating the impact of different learning modalities on learner performance. During the pandemic, learners were assigned to either modular learning (printed modules delivered weekly) or online learning (via Google Classroom and Zoom). Now that face-to-face classes have resumed, the school wants to understand how these modalities affected academic outcomes. The Math department selects a random sample of learners from each group and compares their final exam scores in General Mathematics. <table><tr><td>Modality</td><td>Mean score</td><td>Population standard deviation</td><td>Sample size</td></tr><tr><td>Online learning</td><td>82</td><td>6</td><td>40</td></tr><tr><td>Modular learning</td><td>78</td><td>5</td><td>40</td></tr></table>	Modality	Mean score	Population standard deviation	Sample size	Online learning	82	6	40	Modular learning	78	5	40
Modality	Mean score	Population standard deviation	Sample size										
Online learning	82	6	40										
Modular learning	78	5	40										

	<i>Question:</i> Is there statistical evidence that online learning led to higher academic performance? Use a 5% level of significance, assuming known population standard deviations. Smarter Together: Does Group Work Boost Math Scores? <i>Scenario:</i> Two Grade 11 sections at Sipnayan National High School prepare for a quarterly math quiz. The school’s math coordinator, Ma’am Masagana, wants to explore how different study strategies affect performance. Section Maramihan reviews through group work, solving problems collaboratively and explaining concepts to one another. While Section Mapag-isa reviews through individual study, using self-paced modules and personal notes. After the quiz, Ma’am Masagana collects the following data: <table><tr><th>Section / Method</th><th>Mean score</th><th>Sample standard deviation</th><th>Sample size</th></tr><tr><td>Maramihan (group work)</td><td>84</td><td>5.5</td><td>15</td></tr><tr><td>Mapag-isa (individual study)</td><td>80</td><td>6</td><td>15</td></tr></table> <i>Question:</i> Is there statistical evidence that collaborative learning leads to significantly higher math scores? Use a 5% level of significance, assuming equal population variances.	Section / Method	Mean score	Sample standard deviation	Sample size	Maramihan (group work)	84	5.5	15	Mapag-isa (individual study)	80	6	15
Section / Method	Mean score	Sample standard deviation	Sample size										
Maramihan (group work)	84	5.5	15										
Mapag-isa (individual study)	80	6	15										
V. Synthesis:	Answer the following questions: - Both scenarios compare the mean performance of two independent groups. How do the assumptions given (known population standard deviations vs. equal but unknown variances) affect your choice of hypothesis test? Explain why a z-test is appropriate in one case and a t-test in the other. - Suppose the hypothesis test in each scenario leads to rejecting the null hypothesis. How should school administrators interpret these results when making decisions about learning modalities or study strategies? What factors, beyond statistical significance, should they consider before changing policies or practices? - Design a similar two-group study in another real-life school context (e.g., teaching strategies, assessment methods, or wellness programs). Clearly identify the two groups, state the null and alternative hypotheses, choose the appropriate test, and explain how the results could inform real-world decisions.												

I. Activity No. 3	Assessing Learning Outcomes
II. Objectives:	At the end of the activity, students will be able to apply the appropriate hypothesis test for two independent sample means (z-test or t-test), determine whether a statistically significant difference exists between the groups, and draw real-world conclusions based on the findings.
III. Materials Needed:	pen and paper
IV. Instructions:	Solve the following problems as required. A Grade 11 class conducts a mini-research project on screen time. They ask a random sample of learners to report how many hours they spend on screens each day including phones, tablets, and computers. The results are as follows:

Boys: Average screen time = 4.2 hours
Standard deviation = 0.6
Sample size = 25

Girls: Average screen time = 3.8 hours
Standard deviation = 0.5
Sample size = 25

Is there a significant difference in screen time between boys and girls? Use a 5% level of significance.

2. Two Grade 11 sections take the same math quiz. One section attended a review session the day before, while the other did not. After the quiz, the math teacher, Ma'am Sipnayan, collects the following data:

Group	Mean score	Standard deviation	Sample size
With review	18	2	42
Without review	16.5	2.5	42

Did the review session help learners score significantly higher? Use $\alpha = 0.05$.

3. At a certain school, a group of Grade 11 learners volunteered to join a peer mentoring program aimed at improving study habits and academic performance. Each learner recorded their weekly study hours and scores before joining the program and again after four weeks of participation. The pretest and posttest scores of the learners before and after the peer mentoring program are as follows:

Learner	Pretest Score	Posttest Score
1	12	15
2	14	15
3	15	25
4	13	23
5	13	18
6	16	17
7	17	20
8	19	21
9	15	20
10	17	22

Did the peer mentoring program lead to a significant improvement in the scores? Use 0.05 level of significance.

V. Synthesis:

Answer the following questions:

1. How do you decide whether to use a z-test or a t-test when comparing two groups?
2. What does a statistically significant result really tell us about the difference between two groups?
3. Why does it matter to know if your data is paired or unpaired when interpreting the results of a hypothesis test?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	4
Lesson No.:	11.5	Date:	
Lesson Title/ Topic:	Correlation		
Name:		Grade & Section:	

I. Activity No. 1	Practicing Learned Skills																																							
II. Objectives:	At the end of the activity, students will be able to analyze paired data using Pearson product moment correlation coefficient and simple linear regression to determine the strength, significance, and predictive relationship between variables in real-life contexts.																																							
III. Materials Needed:	pen and paper																																							
IV. Instructions:	Read and solve each problem. 1. Suppose Masagana Pahunan, a store owner in a growing municipality, wants to evaluate whether advertising expenses are truly effective in boosting monthly sales revenue. She assumes that higher spending on advertising campaigns (posters, radio ads, and social media promotions) leads to higher sales, but she wants statistical evidence to support this belief. To test her assumption, she records data for twelve consecutive months: <table><tr><td>Month</td><td>Advertising Expenses (in thousands)</td><td>Sales Revenue (in thousands)</td></tr><tr><td>January</td><td>15</td><td>30</td></tr><tr><td>February</td><td>13</td><td>25</td></tr><tr><td>March</td><td>10</td><td>23</td></tr><tr><td>April</td><td>16</td><td>27</td></tr><tr><td>May</td><td>15</td><td>30</td></tr><tr><td>June</td><td>20</td><td>60</td></tr><tr><td>July</td><td>16</td><td>35</td></tr><tr><td>August</td><td>18</td><td>38</td></tr><tr><td>September</td><td>12</td><td>24</td></tr><tr><td>October</td><td>25</td><td>70</td></tr><tr><td>November</td><td>25</td><td>70</td></tr><tr><td>December</td><td>30</td><td>80</td></tr></table> Task: a. Calculate for the Pearson product moment correlation coefficient and interpret the result. b. Test the significance of the correlation coefficient to determine if there is a significant linear relationship between the variables considered. 2. According to the United Nations Children’s Fund (UNICEF, May 2020), hand washing with soap and water on a regular basis is important for disease prevention. To prevent the spread of infectious diseases, homes, schools, and healthcare facilities must have adequate water, sanitation, and hygiene services. In certain parts of the world, low levels of coverage of these basic services represent significant disparities between and within countries, making these populations vulnerable to the pandemic. Suppose Mayumi Kalusugan, a health worker in a municipality, aims to investigate whether regular handwashing with soap and water reduces the risk of illness. To do this, she gathered data from five barangays, recording	Month	Advertising Expenses (in thousands)	Sales Revenue (in thousands)	January	15	30	February	13	25	March	10	23	April	16	27	May	15	30	June	20	60	July	16	35	August	18	38	September	12	24	October	25	70	November	25	70	December	30	80
Month	Advertising Expenses (in thousands)	Sales Revenue (in thousands)																																						
January	15	30																																						
February	13	25																																						
March	10	23																																						
April	16	27																																						
May	15	30																																						
June	20	60																																						
July	16	35																																						
August	18	38																																						
September	12	24																																						
October	25	70																																						
November	25	70																																						
December	30	80																																						

the proportion of residents who wash their hands after using the toilet alongside the percentage of individuals who fall sick.

Barangay	Percentage of residents with frequent hand washing after using the toilet (%)	Percentage of people who get sick (%)
A	86	10
B	67	23
C	54	30
D	73	11
E	45	25
F	32	27
G	76	11

Task:

a. Calculate for the Pearson product moment correlation coefficient and interpret the result.

b. Test the significance of the correlation coefficient to determine if there is a significant linear relationship between the variables considered.

3. The Department of Education's call for quality and liberating education, reinforced by the K-to-12 Basic Education Program, is consistent with the United Nations' Sustainable Development Goal 4, which promotes equitable access to lifelong learning. Responding to this mandate, Matatag National High School (Senior High School) integrates values of time management and academic excellence into its teaching practices. To evaluate how these values translate into learner outcomes, teacher-researchers at the school designed a study to determine whether there is a correlation between the amount of time spent studying and performance in General Mathematics summative assessments among Grade 11 learners. The data are given below.

Learner	Amount of time spent on studying	Summative assessment score in General Mathematics
1	8	35
2	7	32
3	9	41
4	7	33
5	6	31
6	5	29
7	2	18
8	3	18
9	8	37
10	6	30

Task:

a. Calculate for the Pearson product moment correlation coefficient and interpret the result.

b. Test the significance of the correlation coefficient to determine if there is a significant linear relationship between the variables considered.

V. Synthesis:	Answer the following questions: 1. After computing and testing the Pearson correlation coefficients for all three scenarios (advertising and sales, handwashing and illness, study time and math performance), compare the direction and strength of the relationships. Which relationship appears strongest, and what real-world factors might explain the differences among them? 2. Suppose the correlation in each scenario is found to be statistically significant. How should the store owner, health worker, and school administrators use these results in making decisions? Discuss why a strong or significant correlation should be interpreted with caution when forming policies or interventions. 3. Choose one of the three scenarios and propose an improvement to the study design (e.g., additional variables, larger sample size, or different data collection methods). Explain how your proposed change could strengthen the validity of the correlation analysis and lead to more reliable conclusions.
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I. Activity No. 2	Making Real-World Connections																														
II. Objectives:	At the end of the activity, students will be able to analyze paired data using Pearson product moment correlation coefficient and simple linear regression to determine the strength, significance, and predictive relationship between variables in real-life contexts.																														
III. Materials Needed:	pen and paper																														
IV. Instructions:	Read and analyze the problem and answer the questions that follow. Problem: A certain schools division initiated and implemented the PANGKALIKASAN 101 program in support to the National Greening Program of the government. Under this program, it supports the tree planting activity in schools and in the community. This is in line with guidelines of the Department of Education to integrate the <i>gulayan sa paaralan</i>, ecological solid waste management and tree growing and caring as key components to attain the goals of DepEd on poverty reduction, food security, biodiversity conservation and climate change mitigation and adaptation. With the progression of climate change, expanding urban greenery by at least 10% could help reduce the anticipated rise in local temperatures in the coming decades (Gill et al., 2007). Suppose a research team from a certain institution aiming to explore the connection between tree density and the average ocean temperature during the years 2016 to 2024. The planned dataset is presented below. <table><tr><th>Year</th><th>Number of trees (in billions)</th><th>Mean temperature of oceans (°C)</th></tr><tr><td>2016</td><td>11</td><td>24</td></tr><tr><td>2017</td><td>9</td><td>24</td></tr><tr><td>2018</td><td>8</td><td>25</td></tr><tr><td>2019</td><td>7</td><td>26</td></tr><tr><td>2020</td><td>6</td><td>28</td></tr><tr><td>2021</td><td>7</td><td>27</td></tr><tr><td>2022</td><td>6</td><td>29</td></tr><tr><td>2023</td><td>5</td><td>29</td></tr><tr><td>2024</td><td>5</td><td>30</td></tr></table> Answer the following questions:	Year	Number of trees (in billions)	Mean temperature of oceans (°C)	2016	11	24	2017	9	24	2018	8	25	2019	7	26	2020	6	28	2021	7	27	2022	6	29	2023	5	29	2024	5	30
Year	Number of trees (in billions)	Mean temperature of oceans (°C)																													
2016	11	24																													
2017	9	24																													
2018	8	25																													
2019	7	26																													
2020	6	28																													
2021	7	27																													
2022	6	29																													
2023	5	29																													
2024	5	30																													

	1. Calculate the Pearson product moment correlation coefficient r. 2. Is the value of r positive, negative, or close to zero? 3. What does the sign of r tell you about the direction of the relationship? 4. How strong is the relationship? 5. Is the correlation statistically significant at $\alpha = 0.05$? 6. What does this significance tell you about the reliability of the relationship? What cautions should we take when interpreting correlations?
V. Synthesis:	Answer the following questions: 1. After completing the correlation and regression analyses for the three scenarios, compare the direction and strength of the relationships. Which scenario shows the strongest linear relationship, and how do the contexts (academic performance, attendance habits, and pricing strategy) help explain these differences? 2. Each problem requires predicting an outcome using a regression equation. How reliable are these predictions, especially when the given value lies outside or near the edge of the observed data range (e.g., predicting sales at ₱10 per pack)? Explain the limitations of using regression models for decision-making in real-life situations. 3. Suppose you are a teacher, school administrator, or student entrepreneur using the results of these studies. How would you use the correlation and regression findings to guide actions or interventions? Identify one recommendation for each scenario and justify it using the statistical results.

I. Activity No. 3	Assessing Learning Outcomes																						
II. Objectives:	At the end of the activity, students will be able to analyze paired data using Pearson product moment correlation coefficient and simple linear regression to determine the strength, significance, and predictive relationship between variables in real-life contexts.																						
III. Materials Needed:	pen and paper																						
IV. Instructions:	Solve the following problems as required. Show a complete solution to get complete points. 1. Suppose ten (10) Senior High School learners from a local school were randomly chosen, and their final grades in General Mathematics and General Science were recorded. The table reflects their academic performance across these two core subjects. <table><tr><td>Gen. Math (x)</td><td>90</td><td>95</td><td>87</td><td>94</td><td>77</td><td>80</td><td>82</td><td>90</td><td>87</td><td>82</td></tr><tr><td>Gen. Science (y)</td><td>91</td><td>96</td><td>86</td><td>91</td><td>81</td><td>84</td><td>82</td><td>88</td><td>92</td><td>86</td></tr></table> a. Calculate and interpret the Pearson product moment correlation coefficient. b. Test the significance of the correlation coefficient to determine if there is a significant linear relationship between the variables considered. c. Calculate and interpret the slope and y-intercept of the regression line. d. Find the equation of the regression line. What is the expected grade of a learner in General Science if a learner receives a grade of 88 in General Mathematics?	Gen. Math (x)	90	95	87	94	77	80	82	90	87	82	Gen. Science (y)	91	96	86	91	81	84	82	88	92	86
Gen. Math (x)	90	95	87	94	77	80	82	90	87	82													
Gen. Science (y)	91	96	86	91	81	84	82	88	92	86													

2. Suppose a group of academic track learners launched a study to investigate whether repeated absences in online classes in a certain subject have a measurable impact on learners' semestral grades. Their goal is to uncover patterns that might inform smarter academic interventions. The sample data collected are given below:

Habitual non-presence (x)	2	3	5	6	7	9	10
Semestral grade (y)	96	94	94	90	86	84	79

- Determine and interpret the Pearson product moment correlation coefficient.
- Test the significance of the correlation coefficient to determine if there is a significant linear relationship between the variables considered.
- Determine and interpret the slope and y-intercept of the line of best fit.
- Find the equation of the regression line. Predict the semestral grade of a learner in a certain subject if the habitual non-presence is 8 days.

3. A group of curious and enterprising senior high school learners teamed up to sell a reinvented version of polvoron during Entrepreneurship Week. Treating their venture like a mini business experiment, they deliberately changed the price of their product each day for five consecutive days. Their goal is to explore how daily price adjustments might influence their total sales and earnings, turning their sweet treat into a hands-on lesson in pricing strategy. The data obtained are as follows:

Price per pack (x)	15	25	35	22	30
Number of packs sold (y)	45	35	25	28	30

- Calculate and interpret the Pearson product moment correlation coefficient.
- Test the significance of the correlation coefficient to determine if there is a significant linear relationship between the variables considered.
- Compute and interpret the slope and y-intercept of the line of best fit.
- Determine the equation of the regression line. Estimate the number of packs of polvoron to be sold if the price of polvoron is 10 pesos per pack.

V. Synthesis:

Answer the following questions:

- How does the value of Pearson product moment correlation coefficient influence the reliability of predictions made using the regression line?
- In what ways do correlation and regression complement each other when analyzing relationships between two variables?
- Why is it important to test the significance of the correlation coefficient before drawing conclusions about the relationship between variables?

Performance Task

Task Title: "Study Smart: A Statistical Investigation of Study Habits and Academic Performance"

Overview: You are a data analyst tasked with investigating whether there is a significant relationship between learners' study habits (e.g., hours of study per week) and their academic performance (e.g., grades in Mathematics). This inquiry is supported by various research studies including the study of Gumapac et al. (2025), which found a positive correlation between consistent study routines and improved math achievement among Filipino secondary learners. Similarly, Hermano and Gaudicos (2025) examined the predictive relationship between study habits and mathematical achievement among STEM learners, highlighting how learning styles (visual, auditory, tactile) and study preferences influence performance. Also, Zubair et al. (2024) revealed that learners who studied more than six hours per week consistently outperformed those with fewer study hours. Your task is to collect data, analyze using appropriate statistical tools, and draw conclusions based on your findings and reflect on whether your results support or verify the claims.

References: DOI: <https://doi.org/10.54660/IJMRGE.2024.5.3.720-728>
Findings: Significant relationship between study habits and math achievement; private unaided school learners performed better. (Zubair, T., et. al., 2024)
DOI: 10.47857/Irjms.2025.V06i03.04913
Findings: Auditory learners benefited most from study habits like time on task and test-taking skills. (Hermano, R. H., Gaudicos, AG. P., 2025)
Gumapac, JC. S., Yulo, J.C., Nanud, MN. D. and Baluyos, J. R. (2025). Learners' Basic Math Skills and Study Habits in Relation to their Performance in Mathematics. *International Journal of Science & Engineering Development Research*, 10(1):a551-a564, January-2025. Retrieved <http://www.ijrti.org/papers/IJRTI2501066.pdf>

Instructions:

Data Collection:

- A. Survey at least 20 learners about their (e.g. weekly study hours) and their latest Mathematics test scores.
- B. Organize the data in a spreadsheet.

Formulate Hypotheses:

- A. Create a research question (e.g., "Is there a significant relationship between study hours and math performance?").
- B. Formulate the null and alternative hypotheses.

Determine Statistical Parameters:

- A. Choose an appropriate level of significance (e.g., 0.05).
- B. Identify possible Type I and Type II errors in your context.

Statistical Testing:

- A. Use statistical software (e.g., Excel, GeoGebra, or any calculator) to perform:
 1. z-test (known and unknown standard deviation)
 2. One-sample t-test
 3. Independent sample t-test (e.g., comparing two groups: those who study <5 hours vs. >5 hours)
 4. Paired sample t-test (e.g., pre-test and post-test scores if applicable)

Correlation and Regression:

- A. Create a scatter plot (e.g. study hours vs. test scores).
- B. Draw the line of best fit.
- C. Calculate Pearson's r and interpret the strength and direction of the relationship.

Conclusion:

- A. Interpret the results of your hypothesis tests and correlation analysis.
- B. Discuss whether your data supports your hypothesis.
- C. Reflect on the implications of your findings for learners and educators.

Output Requirements:

1. A written report (3–5 pages) including:
 - a. Introduction and research question
 - b. Data table
 - c. Hypotheses and statistical analysis
 - d. Graphs and charts
 - e. Interpretation and conclusion
2. A short presentation (5–7 minutes) summarizing your findings.

Group Assigned task:

1. Hours of Study per Day/Week
2. Number of Study Sessions per Week
3. Duration of Each Study Session (in minutes or hours)
4. Sleep Hours per Night
5. Number of Subjects Studied
6. Test-Taking Skills (e.g., preparation skills)
7. Number of Distractions (e.g., phone checks per hour)
8. Time Spent on Social Media per Day
9. Attendance Rate (%)
10. Number of Practice Tests Taken

Performance Task Rubric

Criteria	Description	Points
1. Formulation of Hypotheses	Clearly states null and alternative hypotheses relevant to the research question.	10
2. Identification of Statistical Concepts	Accurately illustrates and explains null/alternative hypotheses, level of significance, and types of error.	10
3. Data Collection & Organization	Gather appropriate data, present it clearly in tables/spreadsheets.	10
4. Application of Statistical Tests	Correctly applies z-tests and t-tests using appropriate tools and interprets results.	15
5. Graphical Representation	Constructs a clear scatter plot and accurately draws the line of best fit.	10
6. Correlation Analysis	Calculates Pearson's r and interprets the strength and direction of the relationship.	10
7. Interpretation and Conclusion	Provides a well-reasoned conclusion based on hypothesis testing and correlation results.	15
8. Report Quality	Organized, clear, and complete report with proper formatting, grammar, and visuals.	10

9. Presentation	Engaging, clear, and concise presentation of findings. Uses visuals effectively.	10
TOTAL		100

Criteria	Excellent (Full Points)	Proficient (80%)	Developing (60%)	Beginning (40%)
1. Hypotheses Formulation	Clear, correct null & alternative hypotheses aligned with the research question.	Minor errors in formulation.	Vague or partially correct hypotheses.	Incorrect or missing hypotheses.
2. Statistical Concepts	Accurately illustrates & explains all 4 concepts (null/alt, significance level, errors).	1–2 minor inaccuracies.	Some concepts are missing or unclear.	Most concepts are missing or incorrect.
3. Data Collection & Organization	Complete, relevant data; well-organized tables/spreadsheets.	Mostly complete; minor issues in organization.	Incomplete or disorganized data.	Data missing or irrelevant.
4. Application of Statistical Tests	All tests applied correctly with accurate interpretation.	Most tests correct minor interpretation errors.	Some tests are misapplied or misinterpreted.	Tests misapplied or not attempted.
5. Graphical Representation	Accurate scatter plot with labels & correct line of best fit.	Minor issues in plot or line.	Plot present but unclear or inaccurate.	Plot missing or incorrect.
6. Correlation Analysis	Pearson's r correctly calculated & interpreted.	Minor errors in calculation or interpretation.	Significant errors in calculation or interpretation.	Missing or incorrect analysis.
7. Interpretation & Conclusion	Insightful, evidence-based conclusions; deep understanding.	Reasonable conclusions; minor gaps.	Weak or unclear conclusions.	Conclusions unsupported or incorrect.
8. Report Quality	Well-organized, clear, error-free, with visuals.	Mostly clear; minor errors.	Some organization or grammar issues.	Disorganized or unclear.
9. Presentation	Clear, engaging, confident, with effective visuals.	Clear with minor issues.	Lacks clarity or engagement.	Unclear or incomplete.